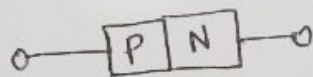


Name: Rakin Afzar

Topic: Semiconductor Diode

A semiconductor diode is a fundamental electronic component that allows current to flow in one direction only. It's typically made of silicon or germanium. The most common type is the P-n Junction diode.

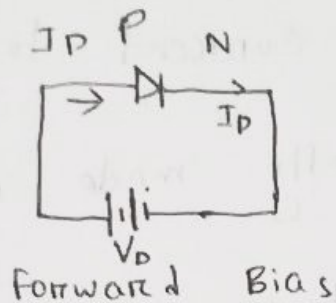


We can apply external voltage. Depending on this, there are three types - such as forward bias, no bias, reverse bias.

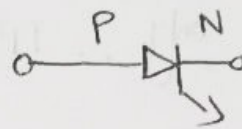
Forward bias:-

When a positive voltage is applied to the P-type material and a negative voltage to the n type material, it forward bias diode.

allowing current to flow easily through it.



Symbol



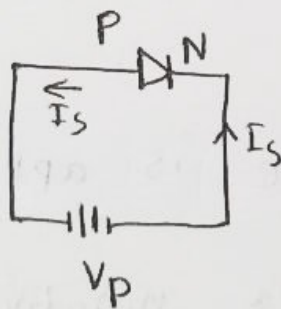
Depletion Layer

$$I_D = I_{maj} - I_s$$

→ reverse saturation

Reverse Bias:-

When a positive voltage is applied to the n-type and a negative voltage is applied to the p-type allowing current to flow easily through it.



Symbol

0

Diode Current - Voltage:

$$I_D = I_S \left(e^{\frac{kV_D}{T_K}} - 1 \right)$$

I_D = Diode Current

I_S = Reverse Saturation

V_D = Voltage across diode

T_K = Temperature in Kelvin.

$$k = \frac{11600}{\eta}$$

η = Ideality factor

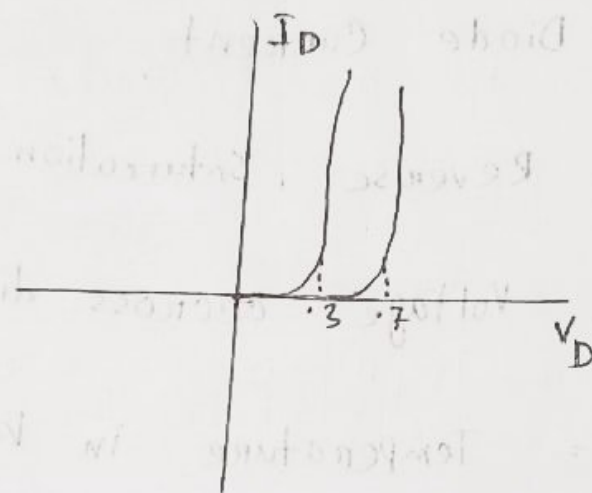
$$\left. \begin{array}{l} \eta = 1 \rightarrow \text{Ge} \\ \eta = 2 \rightarrow \text{Si} \end{array} \right\} \rightarrow I_D \text{ low}$$

$$\eta = 1 \rightarrow \text{Ge, Si} \rightarrow I_D \rightarrow \text{High}$$

$$\frac{k}{T_K} = \frac{1}{\eta V_{th}}$$

Q) V-I characteristics

Forward Bias:-



$V_D \uparrow \uparrow$ $W_d \downarrow \downarrow$ $V_b \downarrow \downarrow \rightarrow$ Barrier Potential
 $\rightarrow$ width of depletion layer

$V_D = V_b \rightarrow$ rapid increase.

No Bias:-

$$I_D = I_S \left(e^{\frac{K_V V_D}{T_K}} - 1 \right)$$

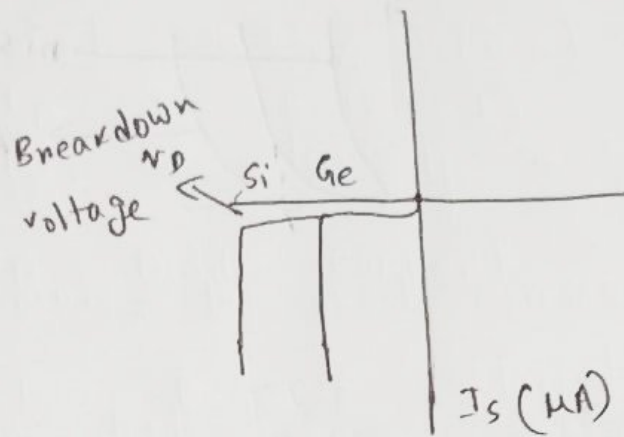
$$V_D = 0$$

$$I_D = I_S (e^0 - 1)$$

$$I_D = I_S (1 - 1)$$

$$I_D = 0$$

Reverse Bias:-



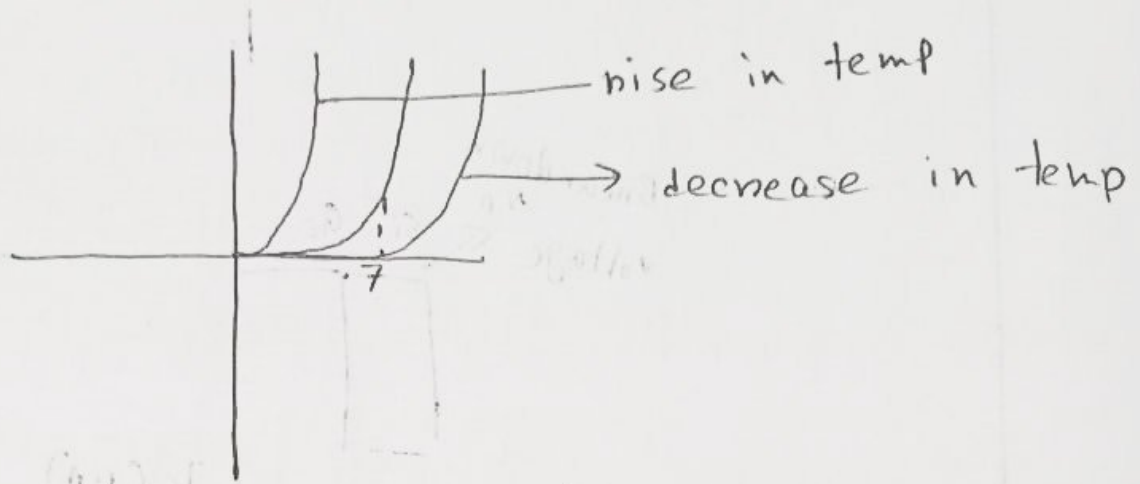
$V_D \uparrow \uparrow$ High kinetic Energy $\rightarrow$ Break covalent bond, which free electron, it's collides with electron and free them. It's a chain reaction.

$$V_D = \text{negative} < 0$$

$$I_D = -I_s.$$

Effects of Temperature

for silicon:-



Forward Bias:-

The graph will shift left 2.5 mV per degree Celsius rise.

The graph will shift right 2.5 mV per degree Celsius ~~rise~~ decrease.

Reverse Bias:-

I_s doubles for every 10°C rise in temperature.

Breakdown voltage $\rightarrow$ increases or decreases depending

on Zener potential.

$$I_s \uparrow \uparrow \uparrow \uparrow$$

Ideal vs Practical diode

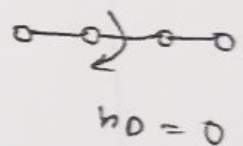
$R_D \rightarrow$ small

forward Bias:- General flow of current = $I_p = I_{max} - I_s$

Reverse Bias: Low flow of current $\rightarrow R_D \rightarrow$ very high

In Ideal,

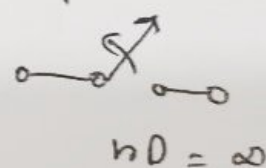
Forward Bias:- No R_D , $r_d = 0$



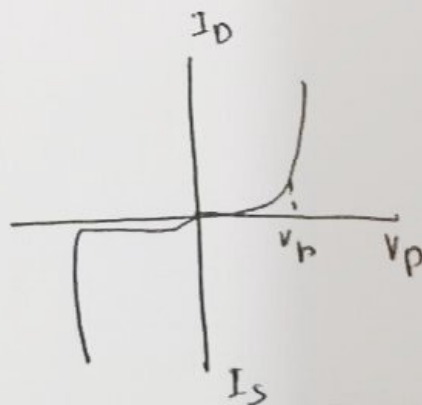
Reverse Bias:-

$$I_s \approx 0$$

$$r_D = \infty$$

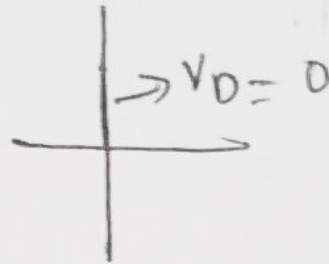


Practical diode:-



Ideal

Forward Bias:-



Reverse Bias:-

$$V_D = \infty$$

$$I_D = \frac{V_D}{\infty}$$

$$I_D = 0$$

